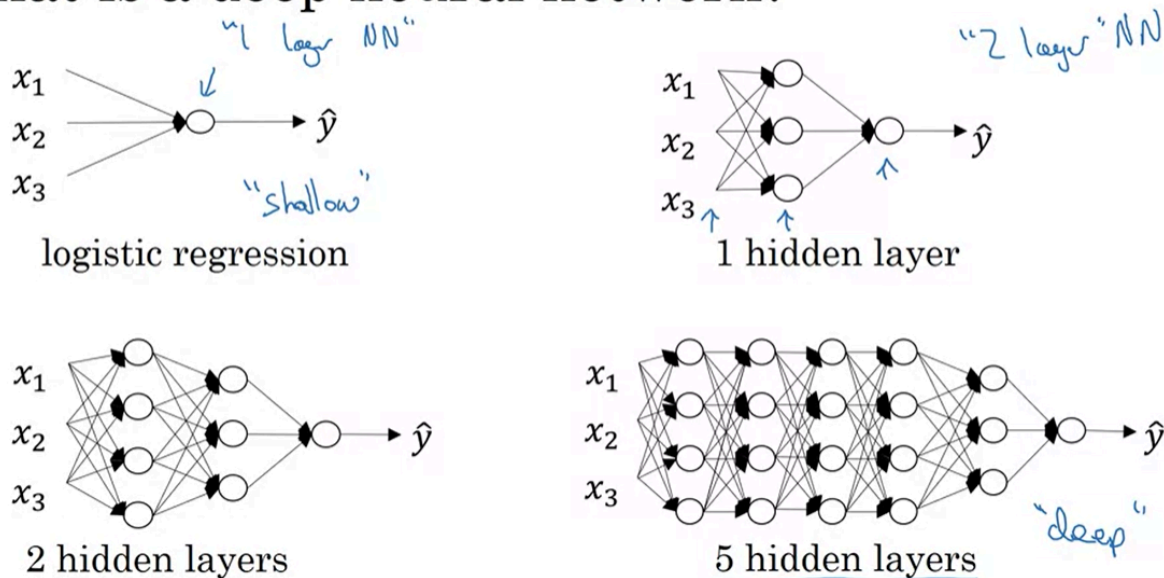


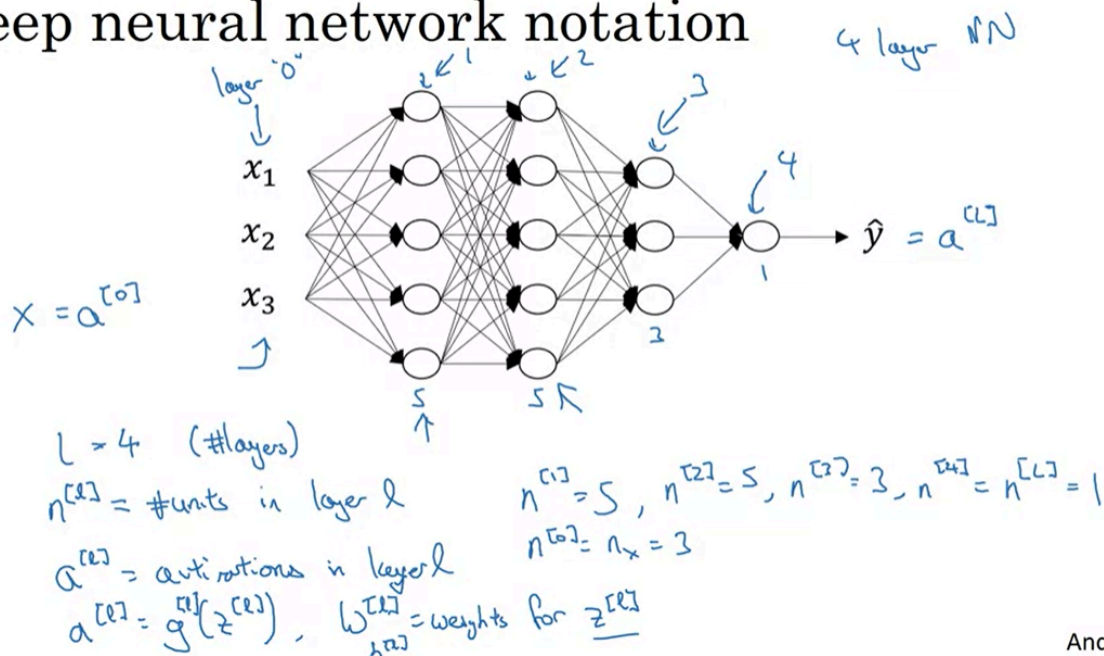
Deep Neural Network

Deep L-layer Neural Network

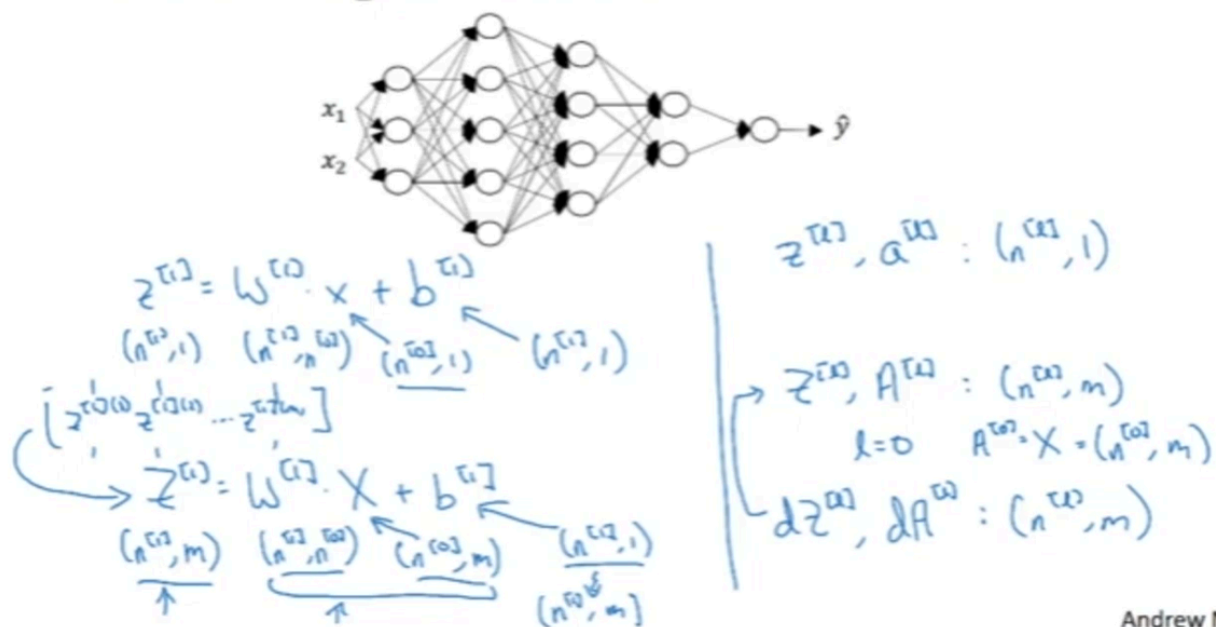
What is a deep neural network?



Deep neural network notation



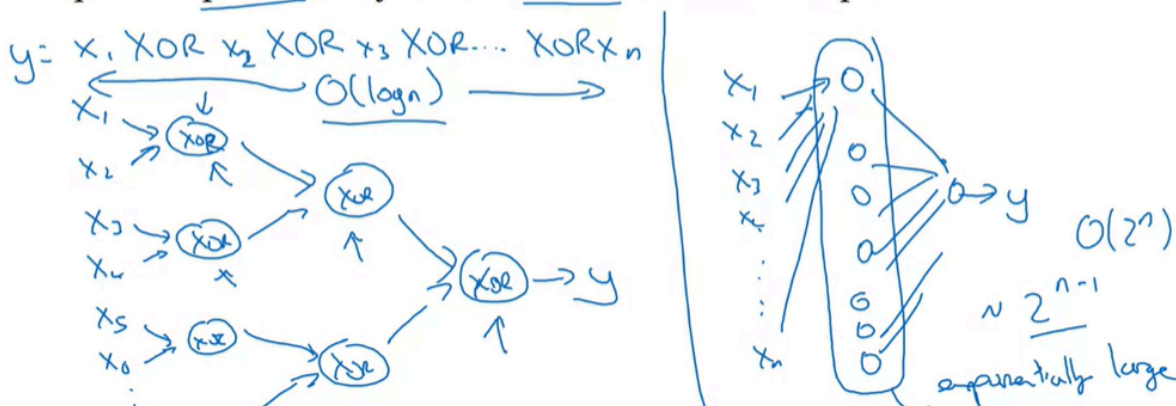
Vectorized implementation



Why Deep Representations?

Circuit theory and deep learning

Informally: There are functions you can compute with a “small” L-layer deep neural network that shallower networks require exponentially more hidden units to compute.



Forward and Backward Propagation

Forward propagation for layer l

→ Input $a^{[l-1]} \leftarrow$

→ Output $a^{[l]}$, cache $(z^{[l]})$

$$z^{[l]} = W^{[l]} \cdot a^{[l-1]} + b^{[l]}$$

$$a^{[l]} = g^{[l]}(z^{[l]})$$

$$\begin{matrix} a^{[l]} \\ A^{[l]} \end{matrix}$$

$$X = A^{[l]} \rightarrow \square \rightarrow \square \rightarrow \square \rightarrow$$

Vertically:

$$z^{[l]} = W^{[l]} \cdot A^{[l-1]} + b^{[l]}$$

$$A^{[l]} = g^{[l]}(z^{[l]})$$

Backward propagation for layer l

→ Input $da^{[l]}$

→ Output $da^{[l-1]}$, $dW^{[l]}$, $db^{[l]}$

$$dz^{[l]} = da^{[l]} * g^{[l]'}(z^{[l]})$$

$$dW^{[l]} = dz^{[l]} \cdot a^{[l-1]T}$$

$$db^{[l]} = dz^{[l]}$$

$$da^{[l-1]} = W^{[l]T} \cdot dz^{[l]}$$

$$dz^{[l+1]} = W^{[l+1]T} dz^{[l]} * g^{[l+1]'}(z^{[l+1]})$$

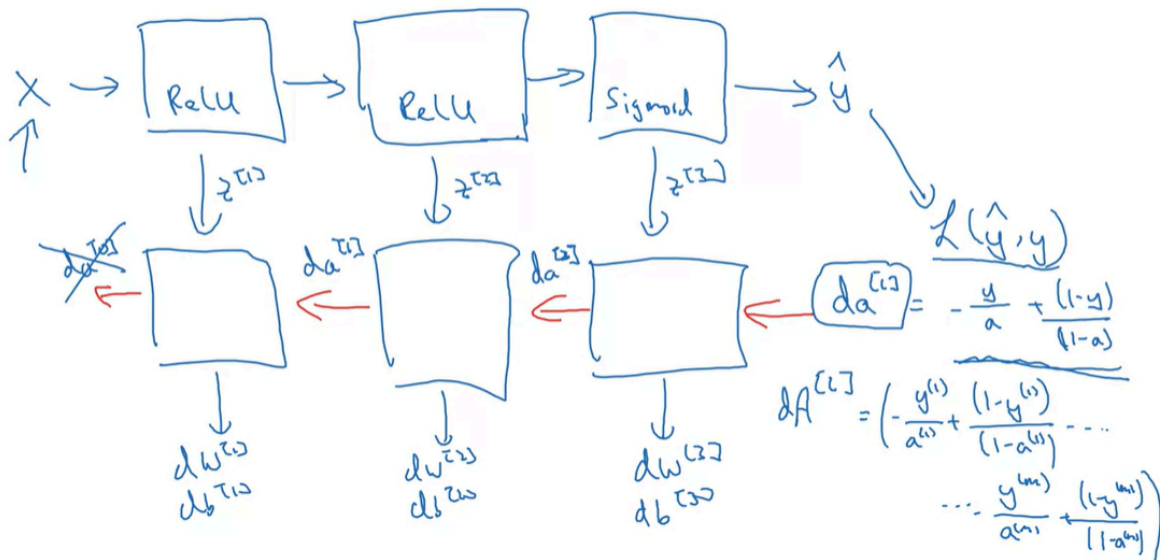
$$dz^{[l]} = dA^{[l]} * g^{[l]'}(z^{[l]})$$

$$dW^{[l]} = \frac{1}{n} dz^{[l]} \cdot A^{[l-1]T}$$

$$db^{[l]} = \frac{1}{n} \text{np.sum}(dz^{[l]}, \text{axis}=1, \text{keepdims}=True)$$

$$dA^{[l-1]} = W^{[l]T} \cdot dz^{[l]}$$

Summary



Note that the formulas shown in the next video have a few typos. Here is the correct set of formulas.

$$dZ^{[L]} = A^{[L]} - Y$$

$$dW^{[L]} = \frac{1}{m} dZ^{[L]} A^{[L-1]T}$$

$$db^{[L]} = \frac{1}{m} \text{np.sum}(dZ^{[L]}, \text{axis} = 1, \text{keepdims} = \text{True})$$

$$dZ^{[L-1]} = W^{[L]T} dZ^{[L]} * g'^{[L-1]}(Z^{[L-1]})$$

Note that $*$ denotes element-wise multiplication)

$\vdots$

$$dZ^{[1]} = W^{[2]T} dZ^{[2]} * g'^{[1]}(Z^{[1]})$$

$$dW^{[1]} = \frac{1}{m} dZ^{[1]} A^{[0]T}$$

Note that $A^{[0]T}$ is another way to denote the input features, which is also written as X^T

$$db^{[1]} = \frac{1}{m} \text{np.sum}(dZ^{[1]}, \text{axis} = 1, \text{keepdims} = \text{True})$$